

Numerical Methods

By

Numerical Methods Group

Numerical Methods are a branch of applied mathematics that deal with finding approximate solutions to mathematical problems that cannot be solved exactly using analytical methods, or whose exact solutions are too complex or time-consuming to compute.

Applications of Numerical Methods

- ✓ Engineering design and simulations
- ✓ Artificial intelligence and machine learning
- ✓ Scientific computing and modeling
- ✓ Economics and financial analysis
- ✓ Medical and biological systems modeling

● Contents of This Course:

- ❑ Chapter (1): Linear System of Equation.
- ❑ Chapter (2): Solution of Algebraic Equations.
- ❑ Chapter (3): Numerical Differentiation and Integration.
- ❑ Chapter (4): Interpolation and Extrapolation.
- ❑ Chapter (5): Curve Fitting, Least-square approximation.
- ❑ Chapter (6): Numerical Solution of Ordinary Differential equations

● Methods of students' assessment:

- ❑ Course Work: 10 Marks
- ❑ Project: 20 Marks
- ❑ Midterm Exam: 20 Marks
- ❑ Final Exam: 50 Marks
- ❑ Total: 100 Marks

● Teaching and Learning Methods :

- ☐ Lectures
- ☐ Tutorial
- ☐ Self-learning
- ☐ Solving problem
- ☐ Discussion

Introduction

● Contents of the Lecture:

- ❑ Sources of Errors
- ❑ Types of Errors
- ❑ Condition Number and Error Behavior

Sources of Errors

When a computational procedure is involved in solving a scientific-mathematical problem, errors often will be involved in the process.

A rough classification of the kinds of original errors that might occur is as follows:

Modelling Errors:

Mathematical modelling is a process when mathematical equations are used to represent a physical system. This modeling introduces errors and are called modelling errors.

Sources of Errors

Blunders and Mistakes:

Blunders are gross errors caused by human actions, not by the numerical method or the computer itself. They occur due to carelessness, misunderstanding, or incorrect data handling.

Mistakes are due to the programming errors.

Sources of Errors

Machine Representation and Arithmetic Errors:

These errors are inevitable when using floating-point arithmetic when using computers or calculators.

Examples are rounding and chopping errors.

Mathematical Approximation Errors:

This error is also known as a truncation error. These errors arise when an approximate formulation is made to a problem that otherwise cannot be solved exactly

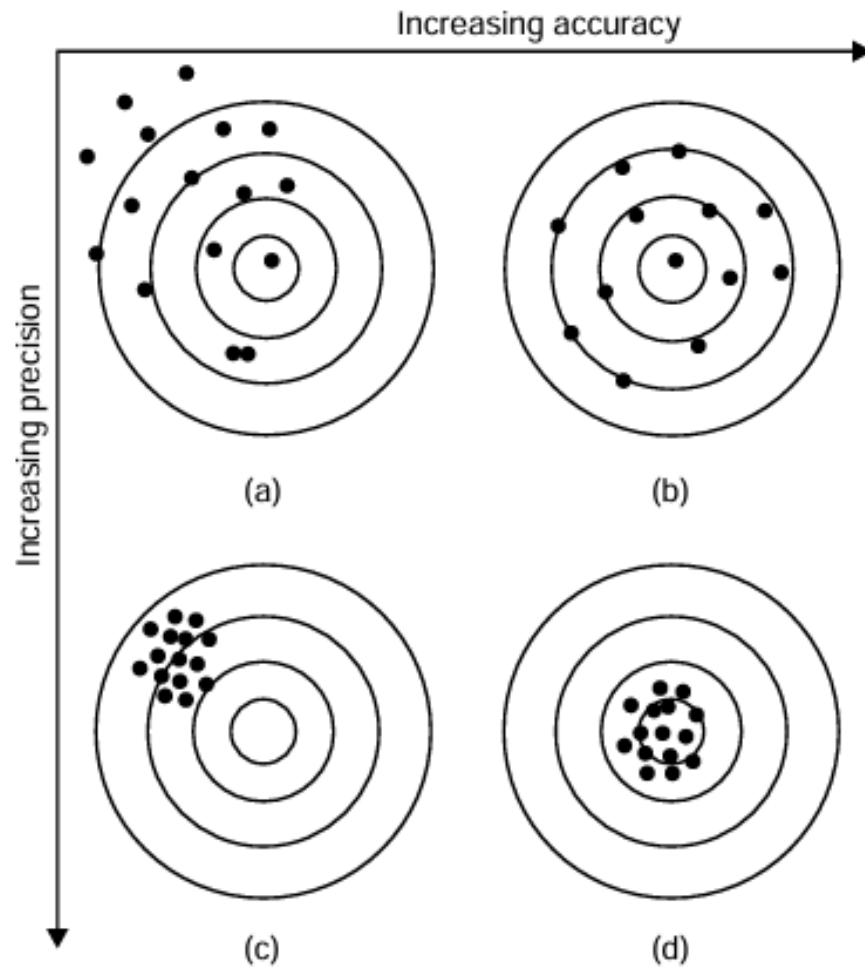
Accuracy and Precision

Accuracy:

refers to how closely a computed or measured value agrees with the true value.

Precision:

refers to how closely individual computed or measured values agree with each other.



Concepts of accuracy and precisions

Inaccuracy and Imprecision

Inaccuracy (also known as bias)

is the systematic deviation from the truth.

Imprecision (uncertainty)

refers to the magnitude of the scatter

Types of Errors

In numerical computations, we come across the following types of errors:

- (a) Absolute and relative errors
- (b) Inherent errors
- (c) Round-off errors
- (d) Truncation errors

- **Absolute error:**

If X_E is the exact or true value of a quantity and X_A is its approximate value, then $|X_E - X_A|$ is called the absolute error E_a .

Therefore absolute error:

$$E_a = |X_E - X_A|$$

- **Relative error:** relative error is defined by

$$E_r = \left| \frac{X_E - X_A}{X_E} \right|$$

- **The percentage relative error:**

$$E_p = 100 E_r = 100 \left| \frac{X_E - X_A}{X_E} \right|$$

- **Significant digits:**

The concept of a significant figure, or digit, has been developed to formally define the reliability of a numerical value. The significant digits of a number are those that can be used with confidence.

If X_E is the exact or true value and X_A is an approximation to X_E , then X_A is said to approximate X_E to t significant digits if t is the largest non-negative integer for which

$$\left| \frac{X_E - X_A}{X_E} \right| < 5 \times 10^{-t}$$

Example :

If $X_E = e$ (base of the natural algorithm = 2.7182818) is approximated by $X_A = 2.71828$, what is the significant number of digits to which X_A approximates X_E ?

Solution :

$$\left| \frac{X_E - X_A}{X_E} \right| = \left| \frac{e - 2.71828}{e} \right| < 5 \times 10^{-6}$$

Hence X_A approximates X_E to 6 significant digits

Example :

Let the exact or true value = $\frac{20}{3}$ and the approximate value = 6.666, what is the absolute error, the relative error and the significant number of digits?

Solution :

The absolute error is

$$E_a = |X_E - X_A| = \left| \frac{20}{3} - 6.666 \right| = 0.000666 = \frac{2}{3000}$$

The relative error is

$$E_r = \left| \frac{X_E - X_A}{X_E} \right| = \left| \frac{\frac{20}{3} - 6.666}{\frac{20}{3}} \right| = \left| \frac{\frac{2}{3000}}{\frac{20}{3}} \right| = \frac{1}{10000}$$

The significant number of digits

$$\left| \frac{X_E - X_A}{X_E} \right| = \left| \frac{\frac{20}{3} - 6.666}{\frac{20}{3}} \right| = \frac{1}{10000} < 5 \times 10^{-4}$$

Hence X_A approximates X_E to 4 significant digits

- **Inherent errors:**

- ❑ Inherent errors are the errors that pre exist in the problem statement itself before its solution is obtained.
- ❑ Inherent errors exist because the data being approximate or due to the limitations of the calculations using digital computers.
- ❑ Inherent errors cannot be completely eliminated but can be minimized if we select better data or by employing high precision computer computations

- **Round-off errors:**

A round-off error is the error that occurs when a number is approximated because a computer can store only a finite number of digits. Since computers cannot represent most real numbers exactly, the stored value is a rounded version of the true value.

- **A chop-off errors:**

A chop-off error occurs when a number is cut off after a fixed number of digits without rounding. The extra digits are simply discarded.

Example :

Given the number π is approximated using $n = 5$ decimal digits.

- 1- Determine the relative error due to chopping and express it as a per cent.
- 2- Determine the relative error due to rounding and express it as a per cent.

Solution :

- 1- The relative error due to chopping is given by

$$E_r(\text{chopping}) = \left| \frac{3.1415 - \pi}{\pi} \right| = 2.949 \times 10^{-5} \text{ or } 0.002949\%$$

- 2- The relative error due to rounding is given by

$$E_r(\text{rounding}) = \left| \frac{3.1416 - \pi}{\pi} \right| = 2.338 \times 10^{-6} \text{ or } 0.0002338\%$$

Example :

If the number $\pi = 4 \tan^{-1}(1)$ is approximated using 5 decimal digits, find the percentage relative error due to,

(a) chopping

(b) rounding.

Solution :

1- Percentage relative error due to chopping is given by

$$E_p(\text{chopping}) = 100 \left| \frac{3.1415 - \pi}{\pi} \right| = 100(2.949 \times 10^{-5})$$

2- Percentage relative error due to rounding is given by

$$E_p(\text{rounding}) = 100 \left| \frac{3.1416 - \pi}{\pi} \right| = 100(2.338 \times 10^{-6})$$

- **Truncation Errors:**

A truncation error occurs when an infinite process is approximated by a finite one. It arises from neglecting higher-order terms in mathematical expressions or stopping an iterative process after a finite number of steps.

■ Condition Number and Error Behavior:

The condition number tells us how much input errors affect the output, independent of the numerical method used.

Consider the first-order Taylor's series of a function given by

$$f(x) = f(a) + f'(a)(x - a)$$

The relative error of $f(x)$ then becomes

$$E_r(f(x)) = \left| \frac{f(x) - f(a)}{f(x)} \right| = \left| \frac{f'(a)(x - a)}{f(a)} \right|$$

The relative error of x becomes

$$E_r(x) = \left| \frac{(x - a)}{a} \right|$$

A condition number is often defined as the ratio of the relative errors given by:

$$\textit{Condition number} = \frac{E_r(f(x))}{E_r(x)} = \left| \frac{a f'(a)}{f(a)} \right|$$

❑ Condition number = 1

The function's relative error equals the relative error in x

(no amplification or attenuation)

❑ Condition number > 1

The relative error is amplified

(small input errors produce larger output errors)

❑ Condition number < 1

The relative error is attenuated

(the function reduces input errors)

Condition number (very large number)

The function is ill-conditioned (highly sensitive to small input errors)

Example :

Compute and interpret the condition number for

1- $f(x) = \sin x$ for $a = 0.51 \pi$

2- $f(x) = \tan x$ for $a = 1.7$

Solution :

1- The condition number is given by:

$$\text{Condition number} = \left| \frac{a f'(a)}{f(a)} \right|$$

for $a = 0.51 \pi$, $f'(a) = \cos(0.51 \pi) = -0.03141$

$f(a) = \sin(0.51 \pi) = 0.99951$

$$\text{Condition number} = \left| \frac{a f'(a)}{f(a)} \right| = \left| \frac{(0.51 \pi) (-0.03141)}{0.99951} \right| = 0.05035$$

Since the condition number is < 1 , we conclude that
the relative error is attenuated.

2- The condition number is given by:

$$\text{Condition number} = \left| \frac{a f'(a)}{f(a)} \right|$$

$$\text{for } a = 1.7, f'(a) = \sec^2(1.7) = 60.2377$$

$$f(a) = \tan(1.7) = -7.6966$$

$$\text{Condition number} = \left| \frac{a f'(a)}{f(a)} \right| = \left| \frac{(1.7)(60.2377)}{-7.6966} \right| = 13.305$$

Thus, the function is ill-conditioned

**THANK YOU
FOR YOUR
ATTENTION**

